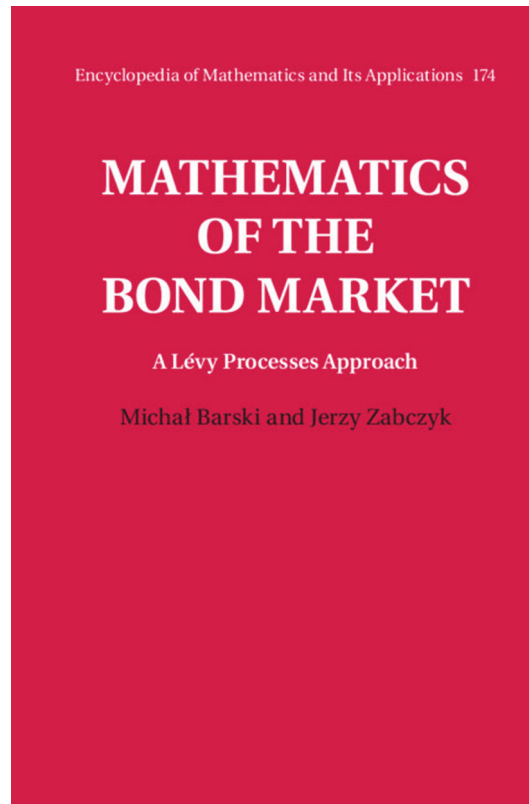


Book review



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Mathematics of the Bond Market: A Lévy Processes Approach, by Michał Barski and Jerzy Zabczyk, Cambridge University Press (2020). Hardback. ISBN 9781107101296.

The book ‘Mathematics of the Bond Market: A Lévy Processes Approach’ by M. Barski and J. Zabczyk delivers what the title promises: It connects two themes that have been central to mathematical finance in the past decades and treats mathematical modelling of fixed-income markets based on Lévy processes in depth. The applications of Lévy processes in mathematical finance go back to the seminal paper of Merton (1976), entitled “Option pricing when underlying stock returns are discontinuous”, which galvanized the research landscape on Lévy models with respect to financial applications. The research on applications of Lévy processes in equity price modelling became particularly active in the 90s, featuring distributions such as VG (Variance Gamma), CGMY (derived from the names of the four co-authors Carr–Geman–Madan–Yor), NIG (Normal Inverse Gaussian), and generalized hyperbolic distributions, whose flexibility yet tractability makes them stand out among mathematical

models even several decades later. Indeed, the activity of the 90s has led to the emergence of several vibrant research communities around the subject: It was catalysed by the paper of Carr and Madan (1999) on option pricing in Lévy models using the fast Fourier transform and other papers written with their co-authors Geman and Yor, as well as by a wealth of contributions on the line of research represented by E. Eberlein and his students from Freiburg, and in the early 2000s by the acclaimed Polish school of researchers, which is well-represented by the authors of this book. Lévy models soon became a part of the standard toolkit of mathematical finance researchers—certainly due also to the excellent textbook of Cont and Tankov (2004), which is the go-to-source for most graduate students who study jump processes.

Interest rate modelling presents a mathematically challenging task, in particular due to the high-dimensionality (even infinite) of the modelling objects. In this sense interest rate models substantially differ from equity price models. Lévy processes have proven to be a very useful modelling tool for this purpose as well: the models driven by Lévy processes are analytically tractable, allow pricing in semi-closed

form based on Fourier transform methods and provide excellent calibration to option market data as they reproduce well the implied volatility surfaces generated by the most liquidly traded options (caps/floors or swaptions). This was shown in several papers by Eberlein and his students, who pioneered research on the topic. However, the existing textbooks typically treat interest rate models based on continuous stochastic processes, apart from a recent book by Eberlein and Kallsen (2019), which dedicates one chapter to interest rate models driven by Lévy processes. The book itself takes a more holistic perspective on mathematical finance in general. This book by M. Barski and J. Zabczyk, is focused entirely on the mathematics of bond markets driven by Lévy processes, and is thus a very welcome contribution to the mathematical finance literature.

Speaking about financial models driven by Lévy processes, it is important to mention that nowadays of course there are also (competitor) models such as models from the rough volatility family which provide an excellent fit to market data throughout a large variety of markets (such as equity, FX, VIX) and settings ((i) derivative pricing, (ii) forecasting, (iii) different data observation frequencies). Though sometimes it is in essence down to the taste of the modeller whether or not one chooses to model the observed phenomena in markets by jump processes or processes with continuous paths, there are aspects that call for modelling using discontinuous processes (such as the presence of jumps in the observed prices, spiky behaviour of interest rates or forward prices in the energy markets). In the very same spirit, we also see the relevance of Lévy models experiencing a third wave in the context of state-of-the-art machine learning models, where analytically tractable benchmarks are needed. The rich and flexible class of Lévy models accommodating numerous examples with well-understood behaviour and easily implementable closed or semi-closed form pricing formulae will certainly be of high value in producing a wide range of market scenarios for benchmarking and testing the performance of algorithms.

Perhaps there is no need to reiterate that M. Barski and J. Zabczyk are widely recognized authors in the field, with a number of papers on the subject of interest rate modelling based on Lévy processes. J. Zabczyk is an expert and co-author of several famous textbooks on stochastic equations in infinite dimensions and on stochastic partial differential equations (SPDEs) with Lévy noise.

The present book is rather theoretical and focuses on the two main issues regarding Lévy-based interest rate models: absence of arbitrage and completeness and the connection to existence and uniqueness of martingale measure(s), with an emphasis on the SPDE point of view of the models (not surprising given the authors' expertise). It does not cover specific examples of models and classes of Lévy processes suitable for applications, or the pricing of interest rate derivatives in this context, but the reader interested in these issues can find the related references in the bibliographic list.

The book is organized into four parts and fifteen chapters and the reader is taken by the hand on a self-contained journey throughout the book.

Parts I and II are dedicated to recalling the basic concepts in the two main areas of focus. In particular, Part I lays out the basic concepts of bond markets in discrete time such as

arbitrage-freeness and completeness and their connection to martingale measures. Random walks are the counterparts of Lévy processes in this setup. Part II recalls the fundamentals of stochastic analysis for Lévy process that will be needed throughout the book.

Part III is the main part of the book and treats the fixed-income markets in continuous time, starting from the main notions and definitions and then presenting the Heath–Jarrow–Morton (HJM) modelling framework driven by Lévy processes. It is worth mentioning that the HJM setup treated here also helps render the book a useful mathematical companion throughout the current LIBOR transition. The Lévy-driven affine short rate models follow and a chapter on the completeness of the presented Lévy term structure models concludes. The final Part IV provides a detailed analysis of the nonlinear stochastic (partial) differential equations arising in the forward rate models. It is a particularly valuable contribution of the book, which is not available elsewhere in this form. The material presented in Parts III and IV is highly sophisticated and mathematically challenging and the reader will appreciate the authors' very pedagogical approach, with numerous examples and comments accompanying the theoretical results.

In conclusion, the book by M. Barski and J. Zabczyk on modelling fixed income markets based on Lévy processes is a textbook that fills perfectly a gap in the existing mathematical finance literature. It serves as an excellent guide to any reader wanting to learn about the mathematics of Lévy processes in modelling the term structure of interest rates. Thanks to its self-contained and carefully balanced approach, no previous knowledge of the theory of fixed income markets is needed. Some minimal knowledge of discontinuous processes would be a plus, but not a necessity. The book is a must-read for entering into this mathematically beautiful and practically highly relevant topic in mathematical finance.

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